RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

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19 October 2026

GitHub: https://github.com/viserion999/LMA_Assignment_1
Code: `RAG_Baseline_vs_Reranking.ipynb`
Outputs: `results/answers.json`, `results/llm_cache.json`

Setup

- **Document:** *Magic: The Gathering Comprehensive Rules* (June 2026, 311 pages). I kept only the numbered rules and split them into 971 chunks (one rule + its sub-rules + section heading; median 187 tokens).
- **Models:** Bi-Encoder `bge-small-en-v1.5` + FAISS exact cosine search; Cross-Encoder `ms-marco-MiniLM-L-6-v2`; LLM `gemini-3.5-flash-lite` (temperature 0, same prompt for both pipelines, told to answer only from the context).
- **Queries:** I wrote 49 player-style questions: negation (14), exception (9), keyword trap (10), buried rule (6), control (10). I fixed the gold rule for each one before running retrieval.

1 Architecture comparison

- **Bi-Encoder:** encodes the question and each chunk *separately* into one vector; relevance is their cosine similarity. Chunk vectors are computed once and indexed, so search is fast. But small words like "can't" or "except" get blurred inside one vector.
- **Cross-Encoder:** reads the question and a chunk *together*; self-attention runs across both texts, so it catches negations and conditions. Nothing can be precomputed: every pair needs its own forward pass.

Pipeline 1: Question → Bi-Encoder + FAISS (top 10) → top 3 → Gemini → Answer 1.
Pipeline 2: Question → Bi-Encoder + FAISS (top 10) → Cross-Encoder re-scores the same 10 → new top 3 → Gemini → Answer 2.

2 Qualitative and rank-shift analysis

Gold chunk in top 3 for both: 31 questions. Fixed by re-ranking: 10. Hurt: 2. Not in top 10 at all: 6.

	Hit@3	MRR@10	Recall@10	Mean gold rank
Pipeline 1 (baseline)	0.673	0.539	0.878	2.61
Pipeline 2 (re-ranked)	**0.837**	**0.710**	0.878	**1.54**

Hit@3 by type: keyword trap 0.70 → 1.00, exception 0.56 → 0.78, buried 0.67 → 0.83, negation 0.57 → 0.71, control 0.90 → 0.90. Recall@10 is unchanged because re-ranking only reorders the same 10 chunks.

2.1 Rank changes for the failure queries

ID	Type	Question (gold rule)	Rank: baseline → re-ranked
Q1	Negation	Can a creature with defender attack? (702.3b)	#5 → #1
Q2	Exception	Can a menace creature be blocked by just one creature? (702.111b)	#5 → #1
Q3	Keyword trap	Can a vigilance creature attack without tapping? (702.20b)	#4 → #1
Q4	Keyword trap	How does the fight keyword action work? (701.14a)	#4 → #1
Q5	Buried	What is the cleanup step used for? (514.1)	#10 → #1
Q6	Exception	Casting a spell without paying its mana cost (118.9)	#4 → #2
Q11	Negation	Is an indestructible creature destroyed by lethal damage? (702.12b)	#4 → #3
Q13	Exception	Two legendary creatures with the same name (704.5j)	#6 → #1
Q27	Keyword trap	If a token creature dies, does it stay in the graveyard? (704.5d)	#10 → #1
Q33	Control	What happens when ward triggers? (702.21a)	#6 → #1
Q46	Control	In what order do spells on the stack resolve? (405.5)	**#1 → #5**
Q49	Exception	Does an Aura stay when the enchanted creature leaves? (303.4c)	**#3 → #4**

- **Why the baseline missed:** it favours chunks that share words with the question. Q1's top 3 were general attack rules; "defender can't attack" was #5. Q27 asks about a token that "dies" in the "graveyard", but the rule says it "ceases to exist" (#10 → #1).
- **Where re-ranking hurt:** Q46 and Q49. The MS MARCO-trained re-ranker preferred an on-topic but less precise rule (e.g. 608.1 over 405.5 for stack order).
- **Not in top 10** (Q24, Q26, Q28, Q30, Q41, Q47): when I ran the Cross-Encoder over all 971 chunks, it found Q26, Q41 at #1 and Q24 at #3, so those misses come from the Bi-Encoder. Q28, Q30, Q47 are missed by both models.

2.2 Answer 1 vs. Answer 2

I generated answers for 23 questions (10 fixed, 2 hurt, 6 not in top 10, 5 controls).

Answer label	Answer 1 (baseline)	Answer 2 (re-ranked)
Correct	6	**14**
Partial	1	2
Refused ("the context does not answer this")	15	7
Wrong	1	**0**

Question	Answer 1 (baseline)	Answer 2 (re-ranked)
Q2 (menace)	"The provided context does not answer this."	"It can't be blocked except by two or more creatures (702.111b)."
Q27 (token)	"The provided context does not answer this."	A token in a zone other than the battlefield ceases to exist.
Q6 (no mana cost)	**Misstated rule:** built the answer on the unpayable-cost rule (118.6a).	Partial: names the right alternative-cost rule (118.9) but leaves out that additional costs still apply.
Q46 (stack)	Correct: the last-added spell or ability resolves first (405.5).	**Omitted clause:** only says the top spell resolves, never "last in, first out".

- **Hallucination:** none. When the key chunk was missing, the baseline refused (15 of 23) instead of inventing a rule. Re-ranking turned 9 refusals into correct answers.
- **Misstated rule:** Q6. Without 118.9 in its top 3, the baseline answered from the wrong rule.
- **Omitted clause:** Q46, but in Answer 2 (lost "last in, first out"). Q49's Answer 2 refused.

3 Latency profiling

I ran everything on CPU only (8 cores). I timed retrieval over 230 runs after a warm-up; Gemini time is one call per query.

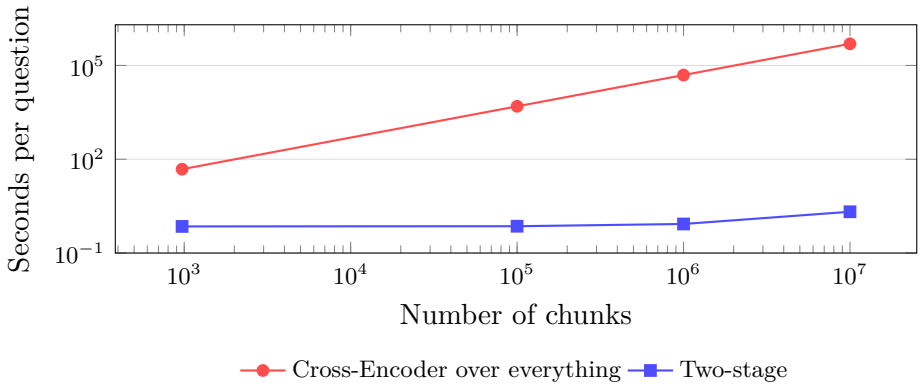
Stage	Median (ms)	Mean (ms)	p95 (ms)
Baseline: Bi-Encoder retrieval, top 10	27.8	28.3	32.0
Re-ranking: Cross-Encoder on 10 pairs	681.7	744.7	1035.2
Pipeline 2 retrieval total	709.1	773.0	1064.0
Gemini generation, Pipeline 1	1137.5	1403.2	1703.6
Gemini generation, Pipeline 2	1073.0	1050.7	1535.2
End-to-end, Pipeline 1	**1165.3**	1431.0	1731.4
End-to-end, Pipeline 2	**1782.1**	1759.8	2244.3

Re-ranking adds ≈0.7 s per query: 25× the baseline search and 38% of Pipeline 2's end-to-end time (CPU cost ≈50–70 ms per pair; much less on a GPU).

4 Scaling question

A Cross-Encoder's score depends on the question, so it cannot be precomputed. Scoring every chunk means N forward passes per query. The cost I measured is linear at ≈49 ms per pair (rows beyond 971 are extrapolated).

Chunks	Cross-Encoder over all chunks	Two-stage (Bi-Encoder + re-rank 10)
971 (measured)	48 s	0.71 s
100,000	4,924 s (1.4 hours)	0.72 s
1,000,000	49,244 s (13.7 hours)	0.85 s
10,000,000	492,444 s (5.7 days)	2.09 s



- The Bi-Encoder does its heavy work once (indexing); each query is one encoding + a fast search.
- So I use two stages: the Bi-Encoder narrows millions of chunks to 10, and the Cross-Encoder only orders those.
- Trade-off: a recall ceiling. If stage one misses the gold chunk (Q24, Q26, Q41), stage two can't recover it.

Limitations: I wrote and judged all 49 questions myself on one document; I corrected the gold lists for Q40 and Q49 after reading the answers; large-scale timings are extrapolated.

Conclusion

Re-ranking raised Hit@3 from 0.67 to 0.84 and correct answers from 6 to 14 (of 23), mostly on negation, exception and keyword-trap questions. It costs about 0.7 s per query on CPU, occasionally demotes the most precise rule, and cannot recover chunks missed by the first stage. The two-stage design gives the best balance of speed and accuracy.